back smoothly. Before it can be added again the violations need to be fixed up.

How can you do this?

Fixing Constraint Violations

When a constraint is added, like `NOT NULL` or `FOREIGN KEY`, all existing rows are checked. Constraint violations prevent their creation. This is true for adding all constraint types.

How can you ensure there are no violations in advance?

To do that, you'll need to query for violations and identify their presence or absence. When dealing with millions of rows it can be a pain to validate all of them.

One technique from the application level would be to iterate through rows in batches and look for problematic values. However, this process can be slow. Querying large data sets is a task PostgreSQL is designed for, and this discovery work can be done in the database.

In the SQL For Devs post "Delete Duplicate Rows",[1] the author describes two methods to identify duplicates and delete them in one query.

In the simpler approach, rows are queried and grouped by the columns that make them unique, and then rows greater than the MIN(id) or less than the MAX(id) are deleted. With the duplicates removed, the constraint can be added.

In the more complex form, a couple of advanced concepts are used to add flexibility over which results to delete. This technique uses a Common Table Expression (`CTE`) for the main query and all rows get a value from the `ROW_NUMBER()` *Window Function*[2] with the `PARTITION BY` option. This partitions (or "segments") the results based on the columns that define uniqueness. CTEs and Window Functions are described only briefly for now.

What's happening is that each row in the results gets a row number. The rows can be ordered in different ways to get a different row number. The rows are deleted based on a row number condition, such as rows numbered > 1.

You've now seen how a multi-column definition of uniqueness can be added to a table, and how to discover duplicates and remove them in advance.

1. https://sqlfordevs.com/delete-duplicate-rows
2. https://www.postgresql.org/docs/current/tutorial-window.html